Query Anchoring Stabilizes Associative Retrieval Without Improving Classification: A Source-Bound Failure Analysis

Prepared for Ryan Gomez

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Abstract

Iterative associative retrieval can converge while degrading, preserving, or failing to improve a classifier. We study a narrow query-anchoring intervention motivated by a retained failure of an experimental EPU digit classifier. The intervention mixes the original query into every recurrent update, yielding a query-conditioned fixed point and a tractable displacement bound. A corrected synthetic successor separates representation from recurrence and records 27,648 example-method outcomes. Multi-prototype representation resolves a constructed multimodal mean bottleneck (accuracy 1.000 versus 0.497), and anchoring reduces mean distractor displacement from 0.783 to 0.077, but anchored recurrence exactly matches its direct same-bank control (0.968). We then run two frozen real-data GPU studies. E6 R2 evaluates grouped cross-validation on 1,797 Optical Digits examples; E7 uses the official writer-independent split with 3,823 training examples from 30 people and 1,797 test examples from a different 13 people. In both studies, anchored recurrence and its direct multi-prototype control make identical predictions at every declared noise level. Utility and robustness effects are exactly zero, noninferiority to logistic regression fails, and only the displacement-reduction hypothesis passes. All retained manifests, source identities, 215,640 raw records, bootstrap intervals, permutation tests, Holm adjustments, and decisions were independently replayed. Query anchoring is therefore a reliable movement constraint in these setups, but the tested recurrence adds no predictive value beyond its representation. This is a source-bound negative efficacy result, not a state-of-the-art or unrestricted generalization claim.

1 Introduction

Associative memories retrieve stored patterns from a query. Classical Hopfield networks formalized content-addressable computation through energy-based dynamics [2]; modern variants use continuous states and softmax retrieval closely related to attention [5]. Convergence is useful only when the resulting state supports the task. A stable fixed point can preserve a wrong label, and a recurrent path can consume substantially more computation than direct retrieval while returning the same decision.

The EPU-QA1 line began from a retained empirical failure in which an iterative classifier underperformed a direct nearest-prototype control. Its original recurrence had no explicit query term in the stationary equation. We therefore ask a bounded question: does retaining the query in every update reduce destructive drift, and does that stability add predictive value over direct inference with the same memory bank?

The study separates two propositions. The mechanistic proposition is that anchoring reduces state displacement. The utility proposition is that this change improves classification beyond a same-bank direct decoder. This distinction is central: stability is neither correctness nor incremental utility.

Our contributions are:

1. a complete contract for the query-anchored recurrence and its direct controls;
2. discovery and repair of a prior implementation defect that invalidated the unanchored synthetic comparison;
3. a factorial synthetic matrix that separates representation, recurrence, sparsity, and full-bank effects;
4. immutable, source-bound synthetic and hosted-GPU records with deterministic metric replay and explicit negative outcomes; and
5. a writer-independent real-data confirmation showing that lower displacement does not improve classification in the tested design.

2 Related work and novelty boundary

Modern Hopfield networks establish continuous softmax retrieval, attention equivalence, and multiple attractor regimes [5]. Universal Hopfield Networks decompose associative retrieval into similarity, separation, and projection,

and show that alternative geometries can materially change performance [4]. Encoded neural representations have also been proposed to improve separability and reduce metastable behavior [3]. These results make broad novelty claims for softmax recurrence, prototype retrieval, fixed context, or representation engineering indefensible here.

The contribution is therefore source-bound. We test whether a fixed query blend repairs a specific failure mode under a declared update. We do not claim that persistent inputs, sparse retrieval, continuous associative memory, or prototype banks are new. We also do not infer biological or hardware relevance.

3 Problem formulation

Let $M \in \mathbb{R}^{k \times d}$ contain memory rows m_i with class labels y_i, and let $q \in \mathbb{R}^d$ be a unit-normalized query. Direct inference selects

$$\hat{\imath}_0 = \arg\max_i \langle m_i, q \rangle, \qquad \hat{y}_0 = y_{\hat{\imath}_0}, \tag{1}$$

with deterministic bank-order tie breaking.

For inverse temperature $\beta > 0$, define

$$w_i(z) = \frac{\exp(\beta \langle m_i, z \rangle)}{\sum_j \exp(\beta \langle m_j, z \rangle)}, \qquad r(z) = \sum_i w_i(z) m_i. \tag{2}$$

The recurrent update is

$$z_{t+1} = \alpha z_t + (1 - \alpha)\left[(1 - \lambda) q + \lambda r(z_t)\right], \qquad z_0 = q. \tag{3}$$

The original rule uses $\lambda = 1$; the query-anchored candidate uses $\lambda = 0.1$. Both use $\alpha = 0.6$, $\beta = 6$, and eight steps. Decoding applies the same bank and argmax rule to z_8.

The synthetic implementation uses unit-normalized rows. Consequently, dot-product and squared-Euclidean nearest-neighbor rankings agree, removing a geometry mismatch between direct and k-nearest-neighbor controls.

4 Conditional analysis

If $D = \max_{i,j} \|m_i - m_j\|$ and $L = \beta D^2/4$, the retrieval Jacobian has operator norm at most L. The recurrent map has Lipschitz constant

$$c \le \alpha + (1 - \alpha)\lambda L. \tag{4}$$

Thus $c < 1$ is a sufficient contraction condition. This establishes a unique fixed point under the bound; it does not establish a correct label.

Because $z_0 = q$ and both q and each retrieved convex combination lie in the unit ball,

$$\|z_t - q\| \leq 2\lambda(1 - \alpha^t). \tag{5}$$

Anchoring therefore directly limits movement. For memories a and b, the score margin obeys

$$\Delta_{t+1} = \alpha\Delta_t + (1 - \alpha)(1 - \lambda)\Delta_0 \tag{6}$$

$$+ (1 - \alpha)\lambda(m_a - m_b)^\top r(z_t). \tag{7}$$

The equation shows why the two propositions differ: a smaller movement can preserve an initial margin without improving whether that margin points to the correct class.

5 Adversarial implementation audit

The prior V4 synthetic runner labeled one method `original_recurrent`, but its unanchored branch removed the query while retaining the candidate's retrieval coefficient 0.1. It therefore computed

$$z_{t+1} = \alpha z_t + (1 - \alpha)0.1r(z_t)$$

rather than the declared original update with $\lambda = 1$. V4 remains historical evidence of the development process, but its comparison involving that method is invalid for the stated original recurrence.

V5 repairs this defect and adds three protections. First, every recurrent method receives an explicit `retrieval_lambda`. Second, direct and recurrent comparisons share the exact same bank. Third, the run manifest binds all checkpoints and all executed source files, while the auditor verifies the full family-seed-example-method Cartesian product and reruns the experiment in a fresh temporary directory.

6 Experimental design

6.1 Known-mechanism families

Each seed creates disjoint training and test draws for two classes.

Unimodal. One mode per class; a class mean is sufficient.

Multimodal. XOR-style class modes share nearly identical class means; a single mean is intentionally non-identifying.

Distractor. Each class has a principal and boundary-adjacent mode, testing sensitivity to bank representation and recurrent movement.

These simulations are constructed diagnostics. They are not handwritten digits and cannot estimate deployment performance.

6.2 Factorial method matrix

The eight methods are direct mean, original mean recurrence, anchored mean recurrence, direct multi-prototype, original top-k recurrence, anchored top-k recurrence, direct full-bank retrieval, and three-neighbor full-bank voting. The mean bank stores one centroid per class; deterministic train-only spherical k-means builds two prototypes per class; the full bank stores every training row. This matrix isolates representation from recurrence. Score-evaluation counts expose computation differences.

6.3 Frozen V5 checks

V5 was designed after V1–V4 were inspected, so it is exploratory even though its four checks were serialized before V5 execution. The checks require: a 0.25 multimodal accuracy gain from multi-prototype over mean direct retrieval; a 0.10 distractor displacement reduction from anchoring; a 0.01 distractor accuracy gain from anchored top-k recurrence over direct multi-prototype retrieval; and a 0.05 distractor accuracy gain from three-neighbor voting over direct multi-prototype retrieval.

Eight fixed random streams are used as simulation replications. We report means, standard deviations, and ranges; they are not treated as human-writer clusters or as a basis for population significance claims.

7 Results

V5 retained 27,648 example-method records. The multi-prototype direct method achieved 1.000 on the constructed multimodal family versus 0.497 for the direct class-mean method, passing the representation check by 0.503. This is expected evidence that the construction actually exposes the intended representation bottleneck.

On the distractor family, the original mean recurrence moved 0.783 on average, whereas the anchored mean recurrence moved 0.077, passing the

Table 1: V5 exploratory synthetic results. Accuracy, mean query displacement, prediction-change rate, and score evaluations are means across eight simulation seeds.

Family	Method	Accuracy	Displacement	Changed	Score evals
distractor	direct_mean	0.959	0.000	0.000	2
distractor	original_mean	0.959	0.783	0.000	18
distractor	anchor_mean	0.959	0.077	0.000	18
distractor	direct_multi	0.968	0.000	0.000	4
distractor	original_topk	0.968	0.241	0.000	36
distractor	anchor_topk	0.968	0.020	0.000	36
distractor	direct_full	0.961	0.000	0.000	96
distractor	knn	0.977	0.000	0.000	96
multimodal	direct_mean	0.497	0.000	0.000	2
multimodal	original_mean	0.497	0.896	0.000	18
multimodal	anchor_mean	0.497	0.087	0.000	18
multimodal	direct_multi	1.000	0.000	0.000	4
multimodal	original_topk	1.000	0.140	0.000	36
multimodal	anchor_topk	1.000	0.013	0.000	36
multimodal	direct_full	1.000	0.000	0.000	96
multimodal	knn	1.000	0.000	0.000	96
unimodal	direct_mean	1.000	0.000	0.000	2
unimodal	original_mean	1.000	0.188	0.000	18
unimodal	anchor_mean	1.000	0.019	0.000	18
unimodal	direct_multi	1.000	0.000	0.000	4
unimodal	original_topk	1.000	0.191	0.000	36
unimodal	anchor_topk	1.000	0.016	0.000	36
unimodal	direct_full	1.000	0.000	0.000	96
unimodal	knn	1.000	0.000	0.000	96

stability check by 0.706. Both produced the same accuracy as direct mean retrieval, 0.959, and neither changed any predicted class.

The central utility screen failed. Anchored top-k recurrence and its direct multi-prototype control both reached 0.968, for an observed difference of exactly 0.000 against a required 0.010. The recurrence used nine times as many score evaluations (36 versus 4). Three-neighbor voting reached 0.977 but improved by only 0.010, failing its deliberately stronger 0.050 screen.

The result is scientifically useful because it discriminates mechanisms. Representation changes solve the deliberately constructed multimodal failure. Anchoring strongly controls trajectory displacement. The tested recurrence adds no classification value beyond its representation.

Table 2: V5 checks fixed before V5 execution. V5 was informed by V1–V4 and remains exploratory.

Check	Required	Observed	Status
REPRESENTATION	0.250	0.503	PASS
ANCHOR-STABILITY	0.100	0.706	PASS
RECURRENT-UTILITY	0.010	0.000	FAIL
FULL-BANK	0.050	0.010	FAIL

8 Real-data GPU studies

8.1 Frozen protocols

E6 R2 and E7 are separately versioned successors to the protected EPU-QA1 and E5-R1 studies; they do not alter or overwrite those historical outcomes. E6 R2 uses the scikit-learn 1,797-row Optical Digits subset, 20 repeats of five-fold grouped stratification over exact-feature groups, and three fixed noise levels. E7 uses the official UCI partition: 3,823 training examples contributed by 30 people and 1,797 test examples from a different 13 people [1]. UCI designates that split as a writer-independent quality measure. The files do not expose per-row writer identities, so E7 establishes partition independence but cannot support writer-cluster-adjusted intervals.

Both studies compare anchored top-k recurrence, original top-k recurrence, direct multi-prototype inference on the identical bank, three-neighbor full-bank voting, and logistic regression. E6 fits inside each training fold; E7 fits only on the official training partition. Both use 20 fixed noise draws at standard deviations 0, 0.15, and 0.30. Inference averages repeated predictions within exact-feature groups, uses 10,000 paired bootstrap draws and 100,000 sign permutations, applies Holm adjustment to the three accuracy hypotheses, tests the declared -0.01 noninferiority margin, and includes a shuffled-label negative control. Frozen CUDA admission requires a GPU with at least 8 GiB memory.

8.2 Hosted execution and results

Both private Kaggle Version 1 notebooks completed on a Tesla T4 with CUDA 12.8 and 15,636,037,632 bytes of device memory. Each retained 107,820 raw example-method-noise records. E6 retained 1,500 fold-metric rows; E7 retained 300 fixed-partition metric rows. Independent local verification checked the downloaded archive against the output directory, source and

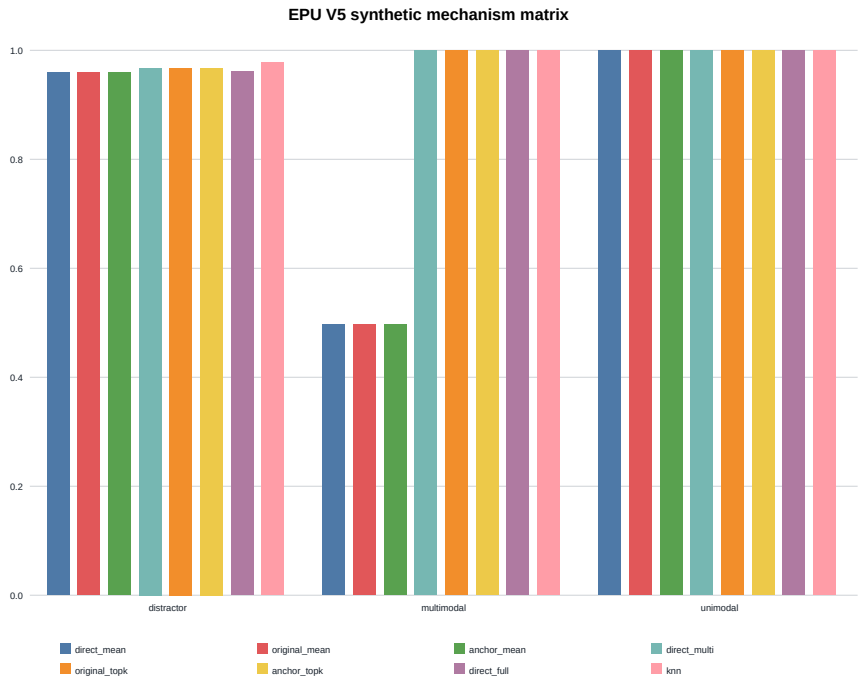


Figure 1: Mean accuracy across eight V5 simulation seeds. Recurrent variants do not exceed their same-bank direct controls in these families.

protocol hashes, raw key completeness, recomputed metrics, exact-feature-group effects, bootstrap intervals, permutation p values, Holm adjustments, hypothesis decisions, negative controls, and completion logs.

In E6, anchored recurrence, direct multi-prototype inference, and original recurrence all achieved accuracies 0.524, 0.507, and 0.464 as noise increased. Full-bank three-neighbor voting achieved 0.986, 0.980, and 0.944; logistic regression achieved 0.968, 0.660, and 0.356. The utility and robustness contrasts were exactly zero. The noninferiority effect was -0.163 with 95% CI $[-0.180, -0.147]$, below the frozen margin. Anchoring reduced displacement by 0.657 with 95% CI $[0.654, 0.660]$.

E7 confirms the same pattern on the official writer-independent split. Anchored recurrence, direct multi-prototype inference, and original recurrence all achieved 0.432, 0.420, and 0.377. Three-neighbor voting achieved 0.978, 0.970, and 0.928; logistic regression achieved 0.950, 0.686, and 0.384. Utility and robustness effects were again exactly zero. The noninferiority effect was -0.263 with 95% CI $[-0.285, -0.242]$. Anchoring reduced displacement by

Table 3: Frozen real-data hypothesis outcomes. Accuracy effects compare anchored top-k recurrence with the stated control. Intervals are 95% exact-feature-group bootstrap intervals. Accuracy-family p values are Holm adjusted.

Study	Hypothesis	Contrast	Effect	CI lower	CI upper	Status
E6	UTILITY	anchor - direct multi	0.000	0.000	0.000	FAIL
E6	ROBUSTNESS	anchor - direct multi at noise 0.30	0.000	0.000	0.000	FAIL
E6	NONINFERIOR-LOGREG	anchor - logistic regression	-0.163	-0.180	-0.147	FAIL
E6	STABILITY	original - anchored displacement	0.657	0.654	0.660	PASS
E7	UTILITY	anchor - direct multi	0.000	0.000	0.000	FAIL
E7	ROBUSTNESS	anchor - direct multi at noise 0.30	0.000	0.000	0.000	FAIL
E7	NONINFERIOR-LOGREG	anchor - logistic regression	-0.263	-0.285	-0.242	FAIL
E7	STABILITY	original - anchored displacement	0.691	0.687	0.694	PASS

Table 4: Aggregate accuracy at zero and maximum noise. The anchored and direct multi-prototype values are identical in both studies at every noise level.

Study	Method	Noise 0	Noise 0.30
E6	anchor_topk	0.524	0.464
E6	knn3	0.986	0.944
E6	logreg	0.968	0.356
E7	anchor_topk	0.432	0.377
E7	knn3	0.978	0.928
E7	logreg	0.950	0.384

0.691 with 95% CI $[0.687, 0.694]$.

Both shuffled-label controls passed the frozen 0.20 ceiling. E7 reproduced exactly on CPU with mean accuracy 0.0879. E6's GPU mean was 0.1119 and its CPU replay mean was 0.1006; individual control values differed although the PASS decision was unchanged. We retain this portability failure and require an independent GPU reproduction before making a cross-runtime reproducibility claim.

9 Threats to validity and limitations

The synthetic generators encode the mechanisms they test. Their positive representation result is therefore a construction check, not a discovery about natural data. V5 was informed by earlier phases and is exploratory. The E6 data do not expose writer identity, and E7 exposes only a writer-independent partition rather than row-level writer labels. Exact-feature grouping pre-

vents duplicate-feature pseudoreplication but cannot replace writer-cluster adjustment. No learned encoder, calibration study, probability model, adversarial perturbation suite, or external acquisition site is evaluated.

The multi-prototype fit uses deterministic spherical k-means and thus adds representational capacity. V5 controls this by comparing direct and recurrent inference on the same bank, but it does not equalize storage with the mean bank. Full-bank methods use more memory. Score evaluations are an implementation-independent cost proxy, not a hardware energy measurement.

The proof bounds are conditional and may be loose. Contraction, energy decrease, and small displacement cannot imply classification accuracy. The present work does not establish architecture novelty, state of the art, population-wide digit efficacy, biological plausibility, or hardware efficiency. E6's cross-runtime negative-control mismatch limits byte-level portability, and neither hosted run is an independent reproduction because the same authors prepared and verified both studies.

10 Reproducibility and evidence integrity

The V5 configuration, raw JSONL records, per-seed CSV, checkpoints, aggregates, hypotheses, report, figure, and audit are retained under a single run directory. Its auditor verifies 27,648 unique Cartesian cells and performs a byte-level deterministic replay. E6 and E7 retain the exact submitted notebooks, protocols, source receipts, Kaggle logs, GPU environment records, compressed raw predictions, fold metrics, hypothesis files, manifests, and evidence archives. Portable verifiers recompute all reported quantities from raw records. The claim-evidence matrix maps each manuscript claim to an artifact and disposition.

This is independent metric verification of hosted evidence, not independent scientific reproduction. A second executor and GPU environment remain required.

11 Discussion

The study illustrates a common evaluation error: an intervention can repair a mathematical property without repairing the task. Query anchoring changes the fixed-point equation and reduces drift, but the direct decoder already contains the relevant decision on these geometries. Repeated retrieval then consumes more score evaluations without adding information. When

the class mean is insufficient, changing representation matters; recurrence over an insufficient bank does not create the missing modes.

The real-data studies now resolve the originally open question. Same-bank direct controls are exactly equal to recurrence, while much simpler established baselines are substantially stronger over most tested conditions. The recurrent utility branch should therefore close for this frozen architecture rather than search adaptively for a favorable configuration. Any successor must state a new mechanism, use a separately versioned protocol, preserve the failed outcomes, and justify additional compute before inspecting protected results.

12 Conclusion

Query anchoring is an effective movement constraint in the tested synthetic and real-data systems, but it does not improve classification over direct inference with the same memory bank. The writer-independent E7 study confirms the negative utility result: anchoring changes trajectories while leaving all declared predictions unchanged. The evidence supports direct inference or the stronger conventional baselines for these tasks and closes the efficacy branch for the frozen architecture. Independent GPU reproduction and writer-cluster-adjusted uncertainty remain open reproducibility questions; they do not change the observed zero utility effect.

References

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A Artifact lineage

The local `arxiv.sty` is a project-specific adaptation of the user's existing local style shim. Only its header text and package date were adapted for this manuscript. The scientific assets are generated only after the V5 adversarial audit passes.

B Disposition vocabulary

Engineering verification means that code and artifact contracts passed. Exploratory synthetic pass or fail refers only to a V5 check. Real-data pass or fail refers to a frozen E6 or E7 decision rule. Writer-independent confirmation describes the official partition, not writer-cluster-adjusted inference. None of these states implies independent reproduction, state-of-the-art performance, submission, or publication.